



EcoSort

Your waste, in the right place.

Multimodal AI

Photo + text

Instant action

University of Cordoba | Neural Networks

The Problem

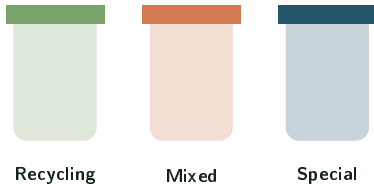
The doubt appears
in seconds

"I have a greasy box"

"Recycle or trash?"

"What about a battery?"

One mistake contaminates



Wrong separation can ruin
an entire recoverable batch.

Target Users



Homes

People who want
to sort waste without
reading long guides.



Schools

Environmental
education with visible
daily decisions.



Cities

Cleaner recycling
streams and fewer
special-waste errors.

The Solution



EcoSort does not only classify.

It tells the user what to do now.

1. Clear photo

2. Short description

3. Verdict + steps

reuse

clean and recycle

compost

special collection point

Functional Demo

	PET	Clean bottle	Dry recycling
	Cardboard	Greasy box	Trash or compost
	Battery	Remote control	Special point
	Clothes	Does not fit	Donate or sell
	Glass	Clean jar	Reuse first

Live demo

API + Expo

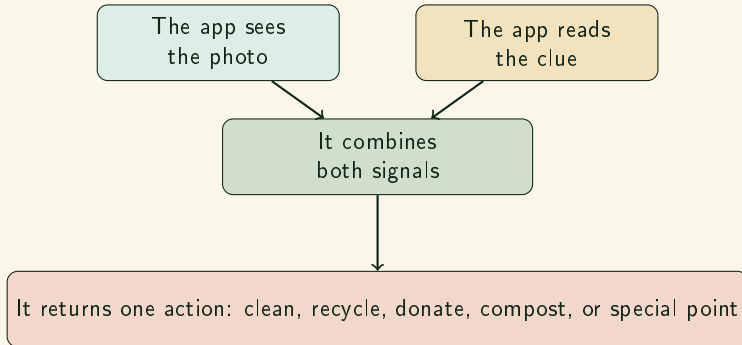
```
scripts/start_api.ps1  
scripts/start_mobile.ps1
```

The app shows a verdict, photo/text evidence, risk level, and disposal steps.

Neural Network Without Jargon

Example

A clean bottle
is not the same as
a chemical container.



Multimodal Architecture

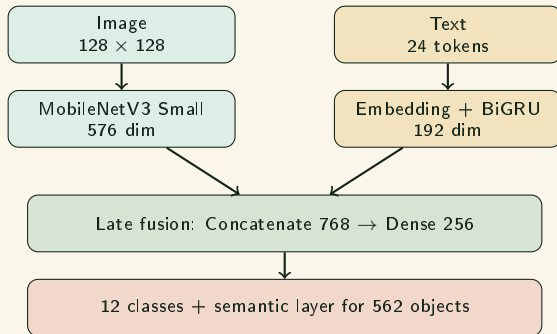
Backend

FastAPI

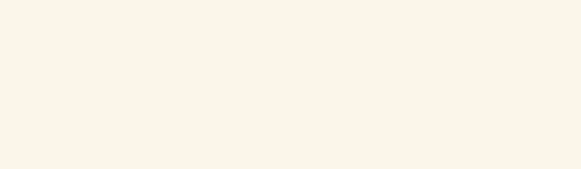
Loads Keras

fuses rules

returns plan

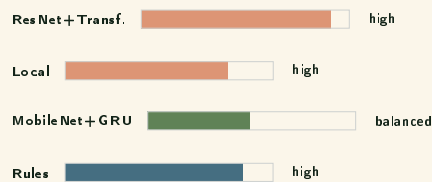


Architectural Decisions

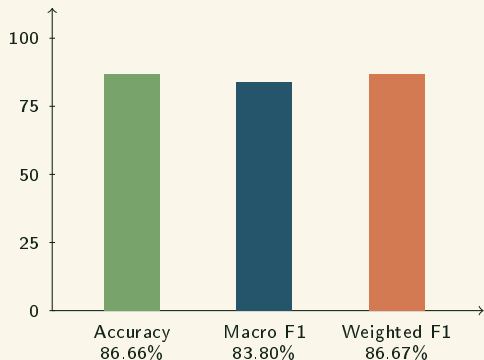


We chose	Because
MobileNetV3	Lightweight for mobile vision.
BiGRU	Enough for short descriptions.
Late fusion	Simpler and explainable.
FastAPI	Loads larger models outside the APK.
Semantic rules	Turn classes into actions.

Architecture comparison



Results



Final model

124 epochs

22,766 test samples

Multimodal Keras

Remote batteries: 92.58%

Dinner leftovers: 90.95%

Artifact: `best_model.keras`

Benefits



Fewer errors

The app explains whether to clean, separate, or reject.



More learning

Users learn at the exact moment of disposal.



More safety

Batteries, chemicals, and broken glass trigger alerts.

Value Proposition and Innovation

From classifying waste to guiding decisions

Product

Photo + text + plan

Impact

Less contamination

Model

Keras + rules

Market

Homes and schools

**EcoSort turns an everyday doubt into the right
environmental action in less than 5 seconds.**



EcoSort

It is not just about throwing away less.
It is about deciding better.

Your waste, in the right place.